No quantum triangular-to-oblique transition in the Gaussian-core crystal: a harmonic first-order instability at or beyond Lindemann melting, removed by the self-consistent harmonic approximation

A preregistered negative result

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Abstract

For the Gaussian-core pair potential $V = \varepsilon e^{-\pi a r^2/l^2}$ at one particle per site, the harmonic energy of a planar Bravais lattice of shape τ is $E = E_{\rm st}(\tau) + \hbar_{\rm eff}\, E_{\rm zp}(\tau)$, with $\hbar_{\rm eff} = \hbar/\sqrt{m\varepsilon l^2}$. We asked (preregistered, Proposal 2 item B1) whether the triangular lattice then undergoes a quantum structural transition to a rhombic/oblique lattice, and whether it comes before melting and survives beyond the harmonic approximation. The preregistered kill criterion reads, verbatim: *"if the transition lies beyond melting or disappears beyond harmonic, report that (negative result)"*. **The criterion is met; this is a negative result. Tier X.** Within the harmonic theory the triangular lattice stops being a local minimum at $\hbar^*_{\rm eff}(a) = -c_{\rm st}/c_{\rm zp}$ for $a \in (0.37668, 1.58148)$ (minimum 0.0877 at $a = 0.5$; 0.1238 at $a = 1$), and the transition is *first order*: a dynamically stable oblique lattice reaches the triangular energy earlier, at $\Lambda_1 \approx 0.75$–$0.78\,\hbar^*_{\rm eff}$ (0.0942 at $a = 1$); for $a = 0.5$–1 the measured slopes at the triangular point on the two inequivalent mirror rays are equal and opposite (ratio -1.02 at $a = 1$), as the cubic invariant allowed by the order-3 stabiliser predicts; the first-order claim itself rests on $\Lambda_1 < \hbar^*_{\rm eff}$. The harmonic Lindemann ratio at the spinodal is at least 0.2798 for every a tested, i.e. at or beyond the top of the quantum-Monte-Carlo band $\gamma_c \in [0.23, 0.28]$; at Λ_1 it is 0.245–0.273 for $a = 0.5$–1 (inside the band, so the Lindemann test alone does not exclude Λ_1 there) and larger elsewhere. The decisive test is anharmonic: in the $T = 0$ self-consistent harmonic approximation (SCHA, two independent implementations agreeing to 8.9×10^{-16}) the triangular lattice stays a local minimum and the lowest stable sampled lattice up to $\hbar_{\rm eff} \approx 0.27$–$0.33$ (the SCHA spinodal, a-dependent), where the SCHA Lindemann ratio is 0.60–0.87, far beyond any melting estimate; the transition disappears. **Caveats.** SCHA is a variational Gaussian theory that omits the second-order cubic $O(\hbar^2)$ term, so this holds *within SCHA*; path-integral Monte Carlo was **not run** (blocked, with numbers); only Bravais lattices were compared; the harmonic Λ_1 winners sit on the harmonic stability edge. **Tier C (quotation-checked literature; no Tier L).** Rydberg dressing realises a soft shoulder whose Fourier transform is negative in places, not a Gaussian; cluster and supersolid physics does not preempt a pure Gaussian (no negative Fourier component); ordinary quantum melting does. The earlier result that the triangular lattice is a local maximum of $E_{\rm zp}$ (doi:10.5281/zenodo.23653833) stands as a statement about the harmonic functional only. A new Rust SCHA module (rusty-SUNDIALS commit `9e85e15`) is part of the work.

Tiers. [A]: kernel-checked Lean (none here). [L]: a theorem-like quotation passed by the Elenchus citation gate (none here, see Section 4). [C]: a human argument, a definition, or a literature statement whose quotation

was checked verbatim against the fetched source text (written [C, quote]); statements taken from a sub-agent's reading and never re-read by us are written [sub] and are in no ledger row. [X]: floating-point exploration, which never supports a claim by itself. Identifiers `B1-X-0001` refer to `ledger/ledger.json` (Section 8).

1 Question, preregistration and what was deviated from

In the prior paper of this programme (P2, doi:10.5281/zenodo.23653833) the harmonic zero-point energy $E_{\rm zp}$ of a unit-covolume planar Bravais lattice was shown numerically *not* to be minimised by the triangular lattice for some completely monotone potentials; for the Gaussian the triangular lattice is a local *maximum* of $E_{\rm zp}$ on $a \in (0.37668, 1.58148)$ [X, from P2]. Since the static energy is minimised by the triangular lattice (Montgomery's theorem, via P2), the total $E_{\rm st} + \hbar_{\rm eff} E_{\rm zp}$ must change its minimiser at some quantum parameter. B1 asks where, and whether it is physical.

Preregistered statements (`results/proposal2/2026-10-26/preregistration.json`, item B1). B1a (X): $\hbar^*_{\rm eff}(a)$ for $a \in (0.377, 1.581)$. B1b: compare with a Lindemann melting estimate (harmonic $\langle u^2 \rangle$). B1c: anharmonic check (SCHA; path-integral Monte Carlo at a few points if feasible). B1d (L): relevance to Rydberg-dressed atoms from verified literature. Priors: B1a 0.85, B1b (transition before melting) 0.4, B1c (survives) 0.25. **Kill:** "if the transition lies beyond melting or disappears beyond harmonic, report that (negative result)".

Deviations and post hoc choices (none changes a prior or the kill condition). (i) The first-order point Λ_1 and its sampled search (Section 3.2) are our definition, not preregistered. (ii) The Lindemann comparison band $\gamma_c \in [0.23, 0.28]$ (QMC true-$\langle u^2 \rangle$ values) was chosen by the producer; we also show the harmonic value 0.30 (Section 3.3). (iii) The continuation-seeded SCHA run beyond the harmonic stability edge, the Rust SCHA module and the ray-slope test were added to close gaps we found. (iv) PIMC was *not run*: blocked (Section 3.5). (v) Only Bravais lattices with one particle per cell are compared. (vi) In B1d the sources we re-read are only four (Section 4).

2 Step-by-step human derivation

Step 0: model and units [C, B1-C-0001]. Particles of mass m on a planar Bravais lattice L_τ of shape $\tau = x + iy$ $(y > 0)$, basis $a_1 = (1/\sqrt{y}, 0)$, $a_2 = (x/\sqrt{y}, \sqrt{y})$, one particle per cell of area $l^2 = 1$. Pair potential $V(r) = \varepsilon\, e^{-\pi a r^2/l^2}$ $(a > 0)$. Lengths in l, energies in ε: $H/\varepsilon = \sum_i \left(-\tfrac{1}{2}\hbar^2_{\rm eff}\nabla_i^2\right) + \sum_{i<j} \tilde{V}$, with $\hbar_{\rm eff} = \hbar/\sqrt{m\varepsilon l^2}$ (the same $\hbar_{\rm eff}$ that appears in P2's functional). In the Gaussian-core-model units $V = \varepsilon e^{-r^2/\sigma^2}$ one has $\rho\sigma^2 = 1/(\pi a)$ and $\hbar/\sqrt{m\varepsilon\sigma^2} = \hbar_{\rm eff}\sqrt{\pi a}$.

Step 1: the harmonic energy, $E = E_{\rm st} + \hbar_{\rm eff} E_{\rm zp}$ [C, B1-C-0001]. Expanding about the lattice sites, the dynamical matrix is $D(\mathbf{k}) = \sum_{R\neq 0}(1 - \cos \mathbf{k}\cdot R)\,\mathrm{Hess}\, V(R)$, with $\mathrm{Hess}\, V(R) = (-2\pi a I + 4\pi^2 a^2 R R^{\mathsf T}) e^{-\pi a|R|^2}$, and each phonon has energy $\hbar_{\rm eff}\sqrt{\lambda_j(D)}$. The ground-state energy per particle is

$$E(\tau) = E_{\rm st}(\tau) + \hbar_{\rm eff}\, E_{\rm zp}(\tau), \qquad E_{\rm st} = \tfrac{1}{2}\sum_{R\neq 0} e^{-\pi a|R|^2}, \qquad E_{\rm zp} = \tfrac{1}{2}\Big\langle \sqrt{\lambda_1(D)} + \sqrt{\lambda_2(D)} \Big\rangle_{\rm BZ}.$$

A shape is *dynamically stable* if $D(\mathbf{k}) \succeq 0$ for all $\mathbf{k}$. $E_{\rm st}(\tau)$ is minimised at $\rho = e^{i\pi/3}$ for every a (Montgomery's theorem, cited second-hand through P2; the original was not opened). This is a statement about shapes at fixed density; it does not include the stress that holds a non-triangular lattice in shape (Section 7).

Step 2: curvature at the triangular point [C, B1-C-0002]. E depends on the lattice only up to rotation, hence $E(\gamma\tau) = E(\tau)$ for $\gamma \in SL_2(\mathbb{Z})$. The map $\gamma(\tau) = (\tau - 1)/\tau$ fixes ρ and has derivative the rotation by $-120°$ there (this is machine-checked in P2, Lean statement `hasDerivAt_gamma`; quoted as prior work, not

re-proved here). A C^2 invariant function then has zero gradient at ρ and a scalar Hessian (a symmetric 2×2 matrix commuting with a 120° rotation is scalar, also checked in P2), so

$$E(\tau)-E(\rho)=c\,\frac{|\tau-\rho|^2}{y_\rho^2}+O(|\tau-\rho|^3),\qquad c=\tfrac14\Delta_{\mathrm h}E(\rho),\quad \Delta_{\mathrm h}=y^2(\partial_x^2+\partial_y^2).$$

Both E_{st} and E_{zp} have this form, with c_{st} and c_{zp}, so $c(\hbar_{\mathrm{eff}})=c_{\mathrm{st}}+\hbar_{\mathrm{eff}}\,c_{\mathrm{zp}}$. The invariance, smoothness and chain-rule steps are not formalised.

Step 3: the harmonic spinodal [C, B1-C-0002; numbers X, B1-X-0001]. Montgomery gives $c_{\mathrm{st}}>0$ and P2 found $c_{\mathrm{zp}}<0$ for $a\in(0.37668,1.58148)$. Then $c(\hbar_{\mathrm{eff}})>0$ exactly when $\hbar_{\mathrm{eff}}<\hbar^*_{\mathrm{eff}}:=-c_{\mathrm{st}}/c_{\mathrm{zp}}$, and the triangular lattice stops being a local minimum at $\hbar^*_{\mathrm{eff}}$. If $c_{\mathrm{zp}}>0$ (outside the interval) there is no such $\hbar^*_{\mathrm{eff}}$. The numbers c_{st} (exact pair sum of $h(u)=(u^2f')'$) and c_{zp} (Richardson stencil, Duffy–Gauss zone rule) are Tier X.

Step 4: why the transition is first order [C, B1-C-0003; test X, B1-X-0003]. Use the disk coordinate $w=(\tau-\rho)/(\tau-\bar\rho)$, in which the order-3 stabiliser of ρ acts as $w\mapsto\omega w$, $\omega=e^{2\pi i/3}$, and the reflection $\tau\mapsto-\bar\tau$ composed with $\tau\mapsto\tau+1$ (every lattice energy is invariant under both) fixes ρ and acts as a reflection of the w-plane; orient the axes so that it is $w\mapsto\bar w$. The stabiliser is thus D_3. A smooth function of $(w,\bar w)$ invariant under D_3 has Taylor expansion in the invariants $|w|^2$, $\operatorname{Re}w^3$, $|w|^4,\dots$: the monomials w^2, $w\bar w^{\,2}$ etc. pick up a factor $\omega^{\pm1,\pm2}\neq1$ and are excluded, but $w^3+\bar w^3$ is allowed. With $w=re^{i\theta}$,

$$\Delta E_{\mathrm{st}}=s_2r^2+s_3r^3\cos3\theta+O(r^4),\qquad \Delta E_{\mathrm{zp}}=z_2r^2+z_3r^3\cos3\theta+O(r^4),\qquad \hbar^*_{\mathrm{eff}}=-s_2/z_2.$$

The *tie coupling* $T(w)=\Delta E_{\mathrm{st}}/(-\Delta E_{\mathrm{zp}})$ is the value of $\hbar_{\mathrm{eff}}$ at which the shape w has the same energy as ρ ($\Delta E_{\mathrm{st}}>0$, $\Delta E_{\mathrm{zp}}<0$). Expanding,

$$T(w)=\hbar^*_{\mathrm{eff}}+\kappa\,r\cos3\theta+O(r^2),\qquad \kappa=\frac{s_3+\hbar^*_{\mathrm{eff}}z_3}{|z_2|}.\tag{1}$$

So if $\kappa\neq0$ a lower-energy shape exists for some $\hbar_{\mathrm{eff}}<\hbar^*_{\mathrm{eff}}$ on every circle around ρ (on the rays with $\kappa\cos3\theta<0$): the triangular lattice is overtaken by a *finite-distortion* state before its quadratic coefficient vanishes, which is a first-order transition (a Landau cubic term). The six rays at $\theta=n\pi/3$ are the three mirror geodesics through ρ; adjacent rays have $\cos3\theta=\pm1$ alternately. The two we scanned, $\operatorname{Re}\tau=\frac12$ upwards and $|\tau|=1$ towards i, differ by 60°, so (1) predicts *equal and opposite* slopes of T in $|w|$. The symmetry argument is Tier C; that $\kappa\neq0$ and the slope ratio are Tier X (Section 3.2).

Step 5: the first-order point [C, B1-C-0003; X]. For a dynamically stable shape τ with $\Delta E_{\mathrm{zp}}<0$, $E(\tau)\le E(\rho)\iff\hbar_{\mathrm{eff}}\ge T(\tau)$. Hence $\Lambda_1(a):=\min_{\tau\ \mathrm{stable}}T(\tau)$ is the smallest coupling at which a stable Bravais competitor reaches the triangular energy. Because we minimise over a finite sample of the stable set, our Λ_1 is an upper bound on the true infimum; because stability is a closed condition, the infimum can sit on the stability boundary, and it does (Section 3.2).

Step 6: the Lindemann criterion and its γ_c^2 sensitivity [C, B1-C-0004]. At $T=0$ in the harmonic theory $\langle|u|^2\rangle/l^2=\hbar_{\mathrm{eff}}\,U(a)$ with $U=\frac12\langle\operatorname{tr}D^{-1/2}\rangle_{\mathrm{BZ}}$ (both Cartesian components, finite in 2D because the $1/|k|$ singularity is integrable). The Lindemann ratio is $\gamma=\sqrt{\langle|u|^2\rangle}/a_{nn}$ with $a_{nn}^2=2/\sqrt3$ for the unit-area triangular lattice. Setting $\gamma=\gamma_c$ gives the harmonic Lindemann coupling

$$\Lambda_L(\gamma_c)=\frac{\gamma_c^2a_{nn}^2}{U(a)}\qquad\Longrightarrow\qquad\frac{\Lambda_1}{\Lambda_L(\gamma_c)}=\Big(\frac{\gamma(\Lambda_1)}{\gamma_c}\Big)^2.$$

Since U is independent of $\hbar_{\mathrm{eff}}$ at fixed a, $\Lambda_L\propto\gamma_c^2$: $\gamma_c\to0.9\gamma_c$ multiplies Λ_L by 0.81 and $\gamma_c\to1.1\gamma_c$ by 1.21. This differs from the Coulomb Wigner crystal of the textbook of this programme (doi:10.5281/zenodo.23295142, [C]), where $\gamma^2\propto r_s^{-1/2}$ and hence $r_s^*\propto\gamma_c^{-4}$ ($0.9^{-4}=1.52$). Lindemann is a rule of thumb, not a theorem; the value of γ_c is the weakest input (Section 3.3).

Table 1: Harmonic results per a (`data/result_numbers.json`). $c_{\rm st}$, $c_{\rm zp}$: curvature coefficients at ρ; $\hbar^*_{\rm eff} = -c_{\rm st}/c_{\rm zp}$; γ: harmonic Lindemann ratio $\sqrt{\langle|u|^2\rangle}/a_{nn}$ at the triangular lattice; Λ_1 and its shape τ: smallest tie coupling of a dynamically stable sampled lattice (Step 5). Tier X.

a	$\rho\sigma^2$	$c_{\rm st}$	$c_{\rm zp}$	$\hbar^*_{\rm eff}$	$\gamma(\hbar^*_{\rm eff})$	Λ_1	$\tau(\Lambda_1)$	$\gamma(\Lambda_1)$
0.3	1.061	0.0017	0.1781			none ($c_{\rm zp} > 0$)		
0.4	0.796	0.0138	−0.0799	0.1732	0.505	0.1307	(0.500, 1.000)	0.438
0.45	0.707	0.0257	−0.2688	0.0956	0.337	0.0746	(0.305, 0.955)	0.298
0.5	0.637	0.0404	−0.4605	0.0877	0.299	0.0683	(0.305, 0.955)	0.264
0.6	0.531	0.0724	−0.7864	0.0921	0.280	0.0714	(0.305, 0.955)	0.247
0.7	0.455	0.0993	−0.9901	0.1003	0.280	0.0769	(0.300, 0.955)	0.245
0.8	0.398	0.1158	−1.0707	0.1082	0.286	0.0826	(0.300, 0.960)	0.250
0.9	0.354	0.1217	−1.0521	0.1157	0.298	0.0883	(0.300, 0.960)	0.260
1	0.318	0.1191	−0.9620	0.1238	0.313	0.0942	(0.300, 0.960)	0.273
1.1	0.289	0.1107	−0.8247	0.1343	0.335	0.1023	(0.300, 0.955)	0.292
1.2	0.265	0.0991	−0.6598	0.1502	0.367	0.1131	(0.300, 0.955)	0.319
1.3	0.245	0.0861	−0.4824	0.1785	0.417	0.1313	(0.460, 1.040)	0.357
1.4	0.227	0.0731	−0.3035	0.2407	0.506	0.1426	(0.485, 1.040)	0.390
1.5	0.212	0.0609	−0.1314	0.4633	0.738	0.1672	(0.500, 1.035)	0.443
2	0.159	0.0202	0.4945			none ($c_{\rm zp} > 0$)		

Step 7: the self-consistent harmonic approximation for Gaussians in closed form [C, B1-C-0005]. The trial state is the ground state of a harmonic crystal with an effective dynamical matrix $D_{\rm eff}(\mathbf{k})$. The displacement difference $w_R = u_R - u_0$ is Gaussian with covariance (mode sum of $\hbar/2\omega$ times $|e^{i\mathbf{k}\cdot R} - 1|^2 = 2(1 - \cos\mathbf{k}\cdot R)$)

$$C_R = \hbar_{\rm eff} \left\langle (1 - \cos\mathbf{k}\cdot R)\, D_{\rm eff}(\mathbf{k})^{-1/2} \right\rangle_{\rm BZ}.$$

For the Gaussian potential every needed average is closed form. With $b = \pi a$, completing the square in $\int d^2w\, \mathcal{N}(w; 0, C)\, e^{-b|R+w|^2}$ (precision C^{-1} plus $2bI$) gives

$$\langle V(R+w)\rangle = \det(I + 2bC)^{-1/2} \exp\!\big(-b\, R^{\mathsf{T}} (I + 2bC)^{-1} R\big), \tag{2}$$

and $\langle \mathrm{Hess}\, V(R+w)\rangle$ is the Hessian of the right side in R. The variational energy per particle is

$$E_{\rm SCHA}[D_{\rm eff}] = \tfrac{\hbar_{\rm eff}}{4} \langle \mathrm{tr}\, D_{\rm eff}^{1/2}\rangle_{\rm BZ} + \tfrac{1}{2} \sum_{R\neq 0} \langle V(R + w_R)\rangle,$$

(the kinetic energy is half the harmonic zero-point energy of $D_{\rm eff}$), and it is stationary where $D_{\rm eff}(\mathbf{k}) = \sum_{R\neq 0} (1 - \cos\mathbf{k}\cdot R) \langle \mathrm{Hess}\, V(R + w_R)\rangle$ (the standard SCHA condition; we do not re-derive it but verify it numerically, Section 3.4), solved by mixing iteration. *Small-$\hbar_{\rm eff}$ limit:* $C_R = O(\hbar_{\rm eff})$ and $\langle V(R+w)\rangle = V(R) + \tfrac{1}{2}\,\mathrm{tr}(C_R\, \mathrm{Hess}\, V(R)) + O(C^2)$, so the potential term contributes a second $\tfrac{\hbar_{\rm eff}}{4}\langle \mathrm{tr}\, D^{1/2}\rangle$ (since $\sum_R (1 - \cos)\mathrm{Hess}\, V = D$): $E_{\rm SCHA} = E_{\rm st} + \hbar_{\rm eff} E_{\rm zp} + O(\hbar^2_{\rm eff})$. The $O(\hbar^2_{\rm eff})$ remainder is what SCHA treats only partially (Section 7).

3 Results

3.1 B1a: the harmonic spinodal $\hbar^*_{\rm eff}(a)$ [X, B1-X-0001]

The triangular lattice stops being a harmonic local minimum at $\hbar^*_{\rm eff}(a)$ for every tested $a \in [0.4, 1.5]$ (Table 1). $\hbar^*_{\rm eff}$ has a minimum 0.0877 at $a = 0.5$ and diverges at both ends of (0.37668, 1.58148) where $c_{\rm zp} \to 0$. Over $a \approx 0.45$–1.3 it lies in 0.088–0.178. Negative controls: $a = 0.3$ and $a = 2$ have $c_{\rm zp} > 0$, hence no $\hbar^*_{\rm eff}$, and no stable sampled shape has $\Delta E_{\rm zp} < 0$ there; $a = 0.378$ (just inside) gives $\hbar^*_{\rm eff} = 2.29$, consistent with the sign change at 0.37668.

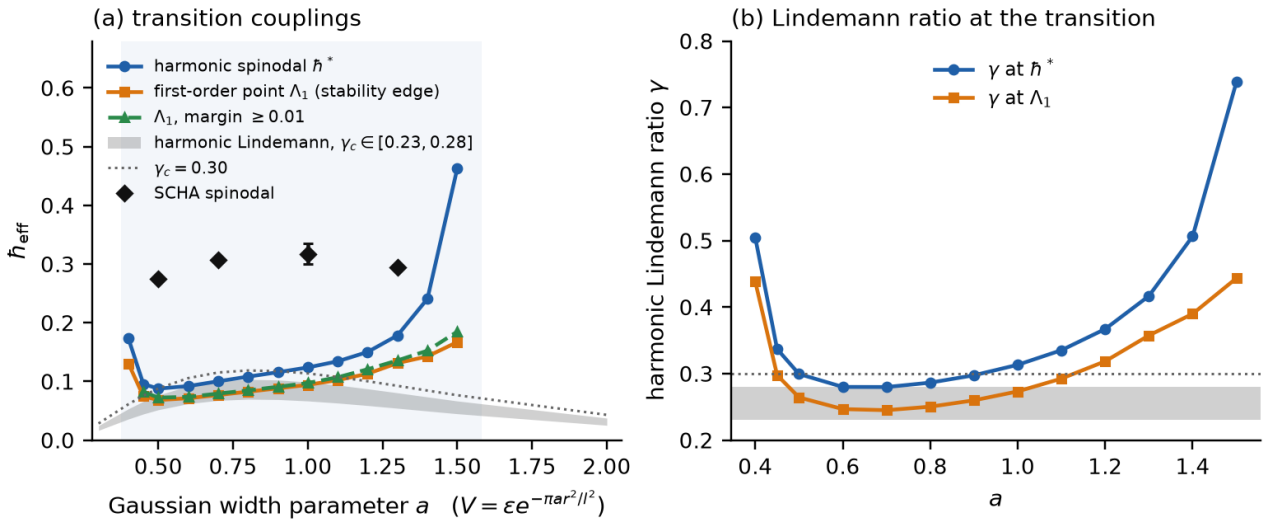


Figure 1: (a) Couplings $\hbar^*_{\mathrm{eff}}$, Λ_1 (stability edge and margin ≥ 0.01), harmonic Lindemann coupling for $\gamma_c \in [0.23, 0.28]$ (shaded) and 0.30 (dotted), and SCHA spinodals (diamonds; at $a = 1$ the interval $(0.300, 0.3347)$). (b) Harmonic Lindemann ratio at $\hbar^*_{\mathrm{eff}}$ and at Λ_1. Tier X.

Table 2: Slope of the tie coupling against $|w|$ on the two mirror rays (quadratic fit, $|w| \leq 0.035$); prediction of Step 4: ratio -1.

a	slope, $\mathrm{Re}\,\tau = \frac{1}{2}$	slope, $\lvert\tau\rvert = 1$	ratio
1	0.0711	-0.0695	-1.023
0.5	0.0955	-0.0945	-1.010
0.7	0.1134	-0.1124	-1.009
0.8	0.1085	-0.1074	-1.010
1.2	-0.0526	0.0553	-0.950

3.2 The transition is first order [X, B1-X-0002, B1-X-0003]

At $a = 1$ the lowest sampled stable competitor appears at $\Lambda_1 = 0.0942 < \hbar^*_{\mathrm{eff}} = 0.1238$, so $\Lambda_1/\hbar^*_{\mathrm{eff}} \in [0.75, 0.78]$ for $a \in [0.45, 1.2]$. Along the ladder in $\hbar_{\mathrm{eff}}$ the argmin over the stable sample is the nearest neighbour of ρ at $\hbar_{\mathrm{eff}} = 0.09$ and jumps to the oblique lattice $0.30 + 0.96i$ at $\hbar_{\mathrm{eff}} = 0.10$ ($\Delta E = -2.86 \times 10^{-4}$ per particle there, from $\Delta E_{\mathrm{st}} = 4.68 \times 10^{-3}$ and $\Delta E_{\mathrm{zp}} = -0.0496$), with no continuous growth of the distortion. The winner is oblique near $\tau \approx 0.30 + 0.96i$ for $0.45 \leq a \leq 1.2$ and near-centred-rectangular ($\mathrm{Re}\,\tau \approx \frac{1}{2}$, $y \approx 1.03$) for $a \geq 1.3$ (Table 1).

The symmetry test. Equation (1) predicts equal and opposite slopes of T against $|w|$ on the rays $\mathrm{Re}\,\tau = \frac{1}{2}$ and $|\tau| = 1$. A quadratic fit over $0.003 \leq |w| \leq 0.035$ gives the slopes in Table 2; the ratio is -1.0 to within 2% for $a = 0.5$–1 (and -0.95 at $a = 1.2$, where both signs are reversed). The opposite signs at the same $|w|$ are what a nonzero D_3-cubic term gives. Because the ray slopes change sign between $a = 1$ and $a = 1.2$, κ must vanish somewhere in between (if the fits are accurate); there the local cubic argument is silent, yet $\Lambda_1/\hbar^*_{\mathrm{eff}}$ stays at 0.75–0.78, so what carries the first-order claim is the finite-distance competitor (Table 1), not the cubic term alone.

The winners sit on the harmonic stability edge. Their stability margin ($\min \lambda/|k|^2$) is 2.3×10^{-5} to 2.4×10^{-3}, against 0.034 for the triangular lattice at $a = 1$. Restricting to shapes with margin ≥ 0.01 moves Λ_1 only by 3–11 % (at $a = 1$: 0.0973), so "first order with $\Lambda_1 \ll \hbar^*_{\mathrm{eff}}$" is robust; but *which* phase lies beyond is not determined by harmonic Bravais theory.

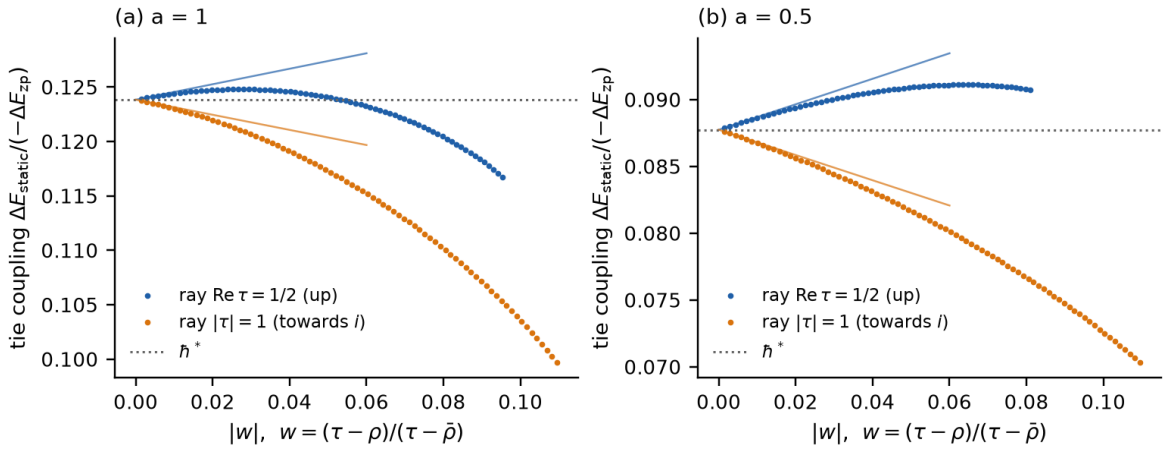


Figure 2: Tie coupling T against $|w|$ along the two mirror rays at $a = 1$ and $a = 0.5$; thin lines: linear parts of the quadratic fits; dotted: $\hbar^*_{\text{eff}}$. Tier X.

Table 3: Harmonic Lindemann couplings $\Lambda_L(\gamma_c)$ and the ratio $\Lambda_1/\Lambda_L = (\gamma(\Lambda_1)/\gamma_c)^2$; ratios < 1: the first-order point comes before melting. Tier X.

a	$\Lambda_L(0.23)$	$\Lambda_L(0.28)$	$\Lambda_L(0.30)$	Λ_1	$\Lambda_1/\Lambda_L(0.23)$	(0.28)	(0.30)
0.4	0.0360	0.0533	0.0612	0.1307	3.63	2.45	2.14
0.45	0.0445	0.0660	0.0758	0.0746	1.68	1.13	0.98
0.5	0.0518	0.0767	0.0881	0.0683	1.32	0.89	0.78
0.6	0.0621	0.0921	0.1057	0.0714	1.15	0.78	0.68
0.7	0.0677	0.1004	0.1152	0.0769	1.13	0.77	0.67
0.8	0.0697	0.1034	0.1187	0.0826	1.18	0.80	0.70
0.9	0.0692	0.1025	0.1177	0.0883	1.28	0.86	0.75
1	0.0668	0.0990	0.1136	0.0942	1.41	0.95	0.83
1.1	0.0632	0.0937	0.1076	0.1023	1.62	1.09	0.95
1.2	0.0590	0.0874	0.1003	0.1131	1.92	1.29	1.13
1.3	0.0544	0.0806	0.0925	0.1313	2.41	1.63	1.42
1.4	0.0496	0.0736	0.0845	0.1426	2.87	1.94	1.69
1.5	0.0450	0.0667	0.0765	0.1672	3.72	2.51	2.18

3.3 B1b: Lindemann comparison [X, B1-X-0004; γ_c anchors C, quote]

The quantum-Monte-Carlo anchors for γ_c in 2D boson crystals are, verbatim from Astrakharchik *et al.* (cond-mat/0608246v2, p. 3, [C, quote, B1-C-0101]): "$\gamma = 0.230(6)$" for dipoles, "0.235(15) for Yukawa bosons", "0.24(1) for Coulomb bosons" and "a hard-disks $\gamma = 0.279$". Their γ is $\sqrt{\langle (r - r_c)^2 \rangle}/a_L$ with a_L "the lattice length"; reading a_L as the nearest-neighbour distance and $\langle (r - r_c)^2 \rangle$ as the total 2D displacement is our assumption ([C, B1-C-0011]); we did not find the definition of a_L in the text. The harmonic value $\gamma = 0.301$ at the Coulomb Wigner transition (textbook, [C]) is a different quantity (harmonic, fermions). We therefore compare with the band $[0.23, 0.28]$ and also show 0.30 (Fig. 1b, Table 3). **Reading.** (a) At the *spinodal* $\hbar^*_{\text{eff}}$ the ratio is ≥ 0.2798 for all a (smallest at $a = 0.7$, on the upper edge of the band, and rising towards both ends, Table 1): the spinodal is at or beyond melting for every a. (b) At the first-order point Λ_1 the ratio is never below 0.245, so $\Lambda_1 > \Lambda_L(0.23)$ for every a (melting first if $\gamma_c = 0.23$); it is inside $[0.23, 0.28]$ for $a = 0.5, 0.6, 0.7, 0.8, 0.9, 1$ and inside $[0.23, 0.30]$ for $a = 0.45, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1$, i.e. $\Lambda_1 \leq \Lambda_L(\gamma_c)$ only for $\gamma_c \geq \gamma(\Lambda_1)$ (between 0.245 and 0.273 for $a = 0.5$–1) (at $a = 1$: $\Lambda_1/\Lambda_L = 1.41$ for $\gamma_c = 0.23$, 0.95 for 0.28). With the γ_c^2 sensitivity of Step 6 (0.81–1.21 for ±10%) this is *inconclusive to adverse*: B1b (prior 0.4) is not supported, and the Lindemann argument alone does not close the case. (c) The anharmonic $\langle u^2 \rangle$ is larger: at $a = 1$, $\Lambda_1 = 0.0942$ gives $\gamma_{\text{SCHA}} = 0.287$ against $\gamma_{\text{harm}} = 0.273$, pushing towards "melted first".

Table 4: SCHA results. Columns 4–8: $\Delta_{\rm h}E_{\rm SCHA}(\rho)$ at $\hbar_{\rm eff} = 0.05, 0.10, 0.15, 0.20, 0.30$; col. 9: harmonic $\Delta_{\rm h}E(\rho) = 4(c_{\rm st} + \hbar_{\rm eff}c_{\rm zp})$ at $\hbar_{\rm eff} = 0.10$; col. 10–11: SCHA spinodal (first sampled coupling with a stable competitor below ρ, or interval at $a = 1$) and its $\gamma_{\rm SCHA}$; col. 12–13: $\gamma_{\rm SCHA}$ at the harmonic Λ_1 and the smallest ΔE (competitor $-\rho$, per particle) among stable sampled shapes there (> 0: triangular lowest). Tier X.

a	$\hbar^*_{\rm eff}$	Λ_1	.05	.10	.15	.20	.30	harm. .10	SCHA spin.	γ	$\gamma(\Lambda_1)$	min ΔE
0.5	0.0877	0.0683	0.103	0.077	0.058	0.037	−0.013	−0.023	0.274	0.60	0.267	1.2×10^{-5}
0.7	0.1003	0.0769	0.238	0.146	0.099	0.069	0.006	0.001	0.308	0.67	0.261	2.3×10^{-5}
1	0.1238	0.0942	0.313	0.193	0.117	0.078	0.008	0.092	(0.30, 0.335)	–	0.287	2.6×10^{-5}
1.3	0.1785	0.1313	0.259	0.174	0.105	0.068	−0.008	0.151	0.294	0.75	0.358	1.6×10^{-5}

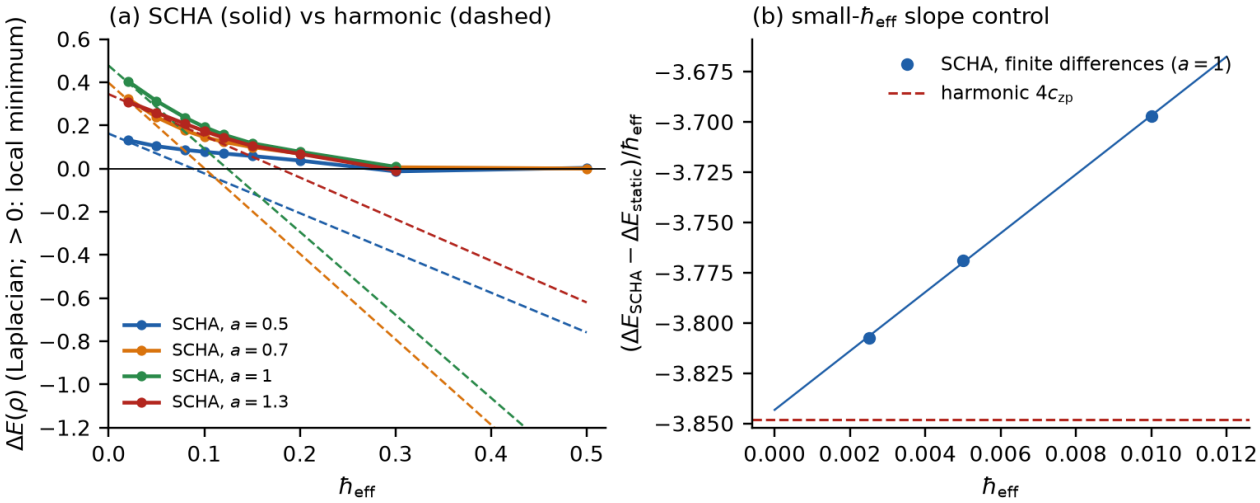


Figure 3: (a) $\Delta_{\rm h}E(\rho)$ against $\hbar_{\rm eff}$: SCHA (solid) and harmonic (dashed). (b) Small-$\hbar_{\rm eff}$ slope control at $a = 1$: $(\Delta_{\rm h}E_{\rm SCHA} - \Delta_{\rm h}E_{\rm st})/\hbar_{\rm eff}$ tends to the harmonic $4c_{\rm zp}$. Tier X.

3.4 B1c: the transition does not survive the self-consistent harmonic approximation [X, B1-X-0005, B1-X-0006]

The SCHA curvature is taken from the SCHA energy of each shape by a stencil, so the full second variation (including the bubble term) is included. Table 4 and Fig. 3a show $\Delta E_{\rm SCHA}(\rho) \equiv \Delta_{\rm h}E_{\rm SCHA}$ (positive: local minimum). At $a = 1$ and $\hbar_{\rm eff} = 0.15$, $\Delta_{\rm h}E_{\rm SCHA} = 0.117$ where the harmonic value is -0.101; at $\hbar_{\rm eff} = 0.300$ it is still $+0.008$. The SCHA spinodal is 0.2745 ($a = 0.5$), 0.3076 ($a = 0.7$), 0.2943 ($a = 1.3$) and in $(0.300, 0.3347)$ at $a = 1$, with SCHA Lindemann ratio 0.60–0.87 (e.g. 0.60 at $a = 0.5$, 0.75 at $a = 1.3$), more than twice γ_c. Near the SCHA breakdown no claim is made: at $a = 1$ the sign of ΔE fluctuates between neighbouring couplings above $\hbar_{\rm eff} \approx 0.33$; at $a = 0.5$, $\Delta_{\rm h}E_{\rm SCHA}(\rho)$ is -0.013 at $\hbar_{\rm eff} = 0.3$ but 0.003 at 0.5 (an apparent re-entrance); the iteration at ρ breaks down at $\hbar_{\rm eff} = 0.617, 0.518, 0.427, 0.370$ for $a = 0.5, 0.7, 1, 1.3$, and there $\gamma_{\rm SCHA} > 0.6$ is outside any crystal description. **Competitors.** At the harmonic Λ_1 every sampled stable competitor (the arc and ray paths to $r = 0.22$ and the harmonic edge winners) lies *above* the triangular lattice in SCHA, by $\geq 10^{-5}$ (last column). A stable competitor below the triangular lattice first appears at the SCHA spinodal itself. **Beyond the harmonic stability edge.** The plain solve is seeded from the bare D and so rejects every harmonically unstable shape by construction. A continuation run seeds each shape with the converged $D_{\rm eff}$ of its neighbour, over 150 shapes per case (the arc $|\tau| = 1$ from ρ to i, $\mathrm{Re}\,\tau = \frac{1}{2}$ up to $y = 1.30$, and a strip $x \in [0.15, 0.30]$ up to $y = 1.12$), at $a = 1$ and 0.5 and $\hbar_{\rm eff} = \Lambda_1, \hbar^*_{\rm eff}, 0.2$ (Table 5): 144–150 are SCHA-stable, of which 103–114 are harmonically unstable (anharmonicity stabilises them), and **none** lies below the triangular lattice. The lowest competitor is the arc point nearest ρ (1.2×10^{-6} to 6.6×10^{-6} above); the lowest shape beyond the harmonic edge is 3.0×10^{-4} to 1.7×10^{-3} above.

Verdict B1c (prior "survives" 0.25): refuted *within SCHA*, which is the kill condition. The negative net result does not rest on SCHA alone: B1b already puts $\hbar^*_{\rm eff}$ at or beyond the Lindemann point.

Table 5: Continuation-seeded SCHA beyond the harmonic stability edge. n: shapes; stable: SCHA-stable; unst.: of these, harmonically unstable; below: SCHA-stable shapes below the triangular energy; min ΔE: smallest over all, and over shapes beyond the harmonic edge. Tier X.

a	$\hbar_{\rm eff}$	$\gamma_{\rm SCHA}(\rho)$	n	stable	unst.	below	min ΔE	beyond edge
1	0.0942	0.29	150	144	103	0	6.6×10^{-6}	1.7×10^{-3}
1	0.1238	0.34	150	150	109	0	4.9×10^{-6}	1.2×10^{-3}
1	0.2000	0.47	150	150	109	0	2.5×10^{-6}	7.0×10^{-4}
0.5	0.0683	0.27	150	150	114	0	3.0×10^{-6}	8.4×10^{-4}
0.5	0.0877	0.30	150	150	114	0	2.7×10^{-6}	7.4×10^{-4}
0.5	0.2000	0.48	150	150	114	0	1.2×10^{-6}	3.0×10^{-4}

Table 6: PIMC feasibility: $a=1$, $\hbar_{\rm eff}=0.10$, competitor $0.30+0.96i$ vs triangular. M: linear cell size, $N=M^2$; shift: finite-size change of ΔE relative to the thermodynamic limit; ΔE_M: finite-cell ΔE; $\hbar\omega_{\rm min}$ of the triangular lattice in the $M\times M$ cell. Tier X.

M	N	shift	ΔE_M	$N\lvert\Delta E_\infty\rvert$	$\hbar\omega_{\rm min}$
6	36	-5.6×10^{-4}	-8.41×10^{-4}	0.010	0.0525
8	64	-2.0×10^{-4}	-4.85×10^{-4}	0.018	0.0401
10	100	-7.0×10^{-5}	-3.56×10^{-4}	0.029	0.0324
12	144	-2.0×10^{-5}	-3.06×10^{-4}	0.041	0.0272
16	256	6.3×10^{-6}	-2.80×10^{-4}	0.073	0.0205
24	576	6.2×10^{-6}	-2.80×10^{-4}	0.165	0.0137

3.5 Path-integral Monte Carlo: BLOCKED [X, B1-X-0009]

No PIMC was run. `code/b1c_pimc_feasibility.py` quantifies why a one- or two-point run within hours on two cores cannot decide the question. (1) A fixed periodic box commensurate with the triangular lattice *imposes* the shape; seeing a distortion needs cell-shape moves at fixed area, which are not implemented. (2) Finite size: an $M\times M$ cell sees only the Γ-centred $M\times M$ $\mathbf{k}$-grid. At $a=1$, $\hbar_{\rm eff}=0.10$, competitor $0.30+0.96i$ the thermodynamic-limit gap is $\Delta E=-2.86\times10^{-4}$ per particle; the finite-size shift of ΔE is -2.0×10^{-4} at $M=8$ and -2.0×10^{-5} at $M=12$ (Table 6), comparable to the signal until $M\gtrsim12$. (3) Thermal smearing: $N|\Delta E|$ is 0.018 ($N=64$) to 0.041 ($N=144$), so $k_BT\ll0.04\,\varepsilon$ is needed for a clear shape preference, while ground-state conditions need $k_BT\ll\hbar\omega_{\rm min}\approx0.027$ at $M=12$, i.e. $\beta\gtrsim250$ with 200+ time slices of 144 particles. (4) Not needed for the verdict: SCHA already shows no instability up to $\gamma\approx0.6$, and at the harmonic Λ_1 ($\gamma\approx0.27$) a PIMC would probe a crystal at the Lindemann threshold, where boson exchange, absent in SCHA, matters too.

3.6 Controls [X, B1-X-0007, B1-X-0008]

Positive. (i) Frozen phonon: the second difference of the explicit supercell pair-sum energy (6×6 cells, all images, V only) equals $\frac{1}{2}e^{\mathsf{T}}D(\mathbf{k})e$ in 12 cases ($a=0.5,1,1.4$; triangular, edge competitor, oblique shape), maximum relative difference 6.3×10^{-6} (the $O(\epsilon^2)$ quartic remainder); a planted bug (dropped Γ-subtraction term of D) gives 0.52, so the control can fail. (ii) P2 reference values are reproduced ($E_{\rm zp}(a=1,\rho)=1.5033523522$, $c_{\rm zp}(1)=-0.96198$, the $c(a)$ table). (iii) Lindemann quadrature: Duffy rule converged to 10^{-13} at $N=16$–128; uniform half-shifted grids converge to it with observed order 1.000 (n^{-1}, from the $1/|k|$ singularity). (iv) SCHA: $\hbar_{\rm eff}\to0$ gives the harmonic energy with remainder $\propto\hbar_{\rm eff}^2$ (coefficient 0.177 at $a=1$; doubling $\hbar_{\rm eff}$ multiplies it by 3.84); the converged $D_{\rm eff}$ is a variational minimum (scaling by 0.97 or 1.03 or interpolating to the bare D raises E by 1.3×10^{-5} to 1.2×10^{-4}); $N=24,32,48$ nodes agree to 3×10^{-11}; the small-$\hbar_{\rm eff}$ slope of the SCHA curvature extrapolates linearly to -3.843 against the harmonic $4c_{\rm zp}=-3.848$ (Fig. 3b; -3.807 to -3.697 at $\hbar_{\rm eff}=0.0025$–0.01), so the stiffening is an $O(\hbar_{\rm eff}^2)$ effect (coefficient ≈14.7) and consistent with harmonic theory at leading order. (v) Python and Rust SCHA agree to 8.9×10^{-16} over 5 shapes (paper-stage rerun, Section 5). Provenance: (i), (iii), the slope in (iv) and (v) are recomputed from the data files or rerun at paper stage; (ii) and the SCHA numbers 0.177, 3.84, 1.3×10^{-5}–1.2×10^{-4} and 3×10^{-11} are quoted from the producer's record (`TRANSITION.md`; the producer's `t_scha.py` was not rerun). **Negative.** $a=0.3$, $a=2$:

no transition (Section 3.1); the square lattice is flagged unstable in both the harmonic and SCHA code; the planted bug above breaks the frozen-phonon match. The square lattice has no SCHA solution at $\hbar_{\rm eff} = 10^{-3}$ when started from the bare D (a seeding statement, not a statement about large $\hbar_{\rm eff}$).

4 Literature (B1d)

What the Tier L gate admits. The Elenchus citation gate accepts as Tier L only a quoted *theorem, lemma, proposition or corollary* and refuses any other kind. We re-read four sources (pdftotext of the arXiv PDFs; hashes in `ledger/sources/MANIFEST.md`), copied each quotation programmatically and ran the gate: the real verdict is `REFUSED` (kind) for every one, and the same gate with the kind relabelled only to test verbatim fidelity accepts all fifteen quotations. We therefore file them as Tier C with the verbatim check, written [C, quote]; there is no Tier L row in this paper. Everything else in this section was read by a literature sub-agent through alphaXiv and never re-read by us ([sub]); it is not in the ledger and should be checked before being relied on.

No quantum Gaussian-core QMC found. Three targeted searches by a sub-agent found no QMC or PIMC study of bosons with a Gaussian pair potential ([sub]; absence by search, not proof of absence). The only quantum result is mean-field: Prestipino, Sergi and Bruno (arXiv:1809.07515v1, [C, quote, B1-C-0102..0105]): p. 11 "The Gaussian repulsion ($n = 2$) is a marginal case: despite there is no evidence of a cluster crystal in two or three dimensions, clear hints of clusterization are detected in one dimension"; p. 15 "for $n = 2$ (where $\tilde{u}(k)$ is everywhere positive) crystallization is swept away at zero temperature"; p. 25 "freezing is completely washed out in the Gaussian, $n = 2$ case". These are mean-field statements, which the authors call "realistic for weak interparticle forces" (p. 25), for a condensate at high density; ours is the opposite regime, one particle per site with $\hbar_{\rm eff} \to 0$. They neither support nor contradict the crystal results here.

No cluster or supersolid preemption of a pure Gaussian [C, B1-C-0007, B1-C-0008]. The two-dimensional Fourier transform of $e^{-\pi a r^2}$ is $a^{-1}e^{-\pi k^2/a} > 0$ for every k (closed form). Reports that cluster crystals and supersolid droplet crystals need a negative Fourier component of V (Cinti–Boninsegni–Pohl arXiv:1402.0840; Boninsegni arXiv:1201.0980 Sec. 2.3; Mendoza-Coto *et al.* arXiv:2010.04631, Sec. III.B, where per the sub-agent the Gaussian ($\alpha = 2$) is exactly the boundary of cluster-forming ability) are [sub]. The weaker statement we rely on, that the Gaussian is the marginal non-clustering case, rests on the verbatim Prestipino quote above. The preemption we do find is ordinary quantum melting (Sections 3.3–3.4).

Classical 2D Gaussian-core model ([sub]): the $T = 0$ ground state is the triangular crystal at any pressure and the highest melting temperature is $\approx 0.0115\,\varepsilon/k_B$ (Prestipino, Saija, Giaquinta, arXiv:1107.0828, Phys. Rev. Lett. 106, 235701), consistent with our static control; melting temperatures read off figures in arXiv:2209.02802 are imprecise and not used.

Rydberg dressing: soft shoulder, not Gaussian [C, quote, B1-C-0009, B1-C-0106..0115; X, B1-X-0010, B1-X-0011]. Zeiher *et al.* (arXiv:1602.06313v3, p. 2) write the dressed interaction as $U_{ij} = U_0/(1 + (R/R_c)^6)$; Henkel, Nath and Pohl (arXiv:1001.3250v2, p. 2) obtain $U(r_{ij}) = \tilde{C}_6/(r_{ij}^6 + R_c^6)$ and state that "due to the inner potential plateau at $r < R_c$, the momentum space potential $\tilde{U}$ has negative attractive contributions around $k = k_{\rm rot} \sim 2\pi/R_c$". We confirm it numerically: the 2D Hankel transform of $1/(1 + (r/R_c)^6)$ has minimum $\tilde{u}/\tilde{u}(0) = -0.0448$ at $kR_c = 4.82$ and is negative on 44% of $0 \le kR_c \le 8$ (two cutoffs agree to 4.9×10^{-11}); the same code reproduces the Gaussian closed form $\pi e^{-k^2/4}$ to 8.9×10^{-16} (Fig. 4). The dressed gas therefore belongs to the cluster/supersolid class, not to the Gaussian class. The reachable quantum parameter $\hbar/\sqrt{mU_0R_c^2}$ (in units of the soft-core radius R_c, not of l): Henkel's footnote 26 gives $\tilde{C}_6 M/(\hbar^2 R_c^4) < 10^{-2}$ for their Fig. 3c, i.e. $\hbar/\sqrt{mU_0R_c^2} > 10$ (using $U_0 = \tilde{C}_6/R_c^6$), deep mean-field. Zeiher's parameters ($U_0/2\pi \approx 1.8$ kHz, $R_c/a_{\rm lat} = 2.02$, $a_{\rm lat} = 532$ nm, rubidium-87, all quoted) give $\hbar^2/(mR_c^2)/h = 100.7$ Hz, $U_0/(\hbar^2/mR_c^2) = 17.9$ and $\hbar/\sqrt{mU_0R_c^2} = 0.237$ (our arithmetic), for atoms pinned in an optical lattice; the measured loss time of the dressed atoms is 130(20) μs (against 2.2 ms expected and the 27 μs bare Rydberg lifetime), far below the motional time $\approx 1/(100.7\ {\rm Hz}) \approx 10$ ms. The soft shoulder has no Gaussian width a, so these numbers cannot be compared with our window $\hbar_{\rm eff} \approx 0.07$–$0.18$ ($a \approx 0.45$–1.3); none is a demonstrated itinerant realisation of that window. A true Gaussian interaction was not found in cold-atom or molecular platforms ([sub]).

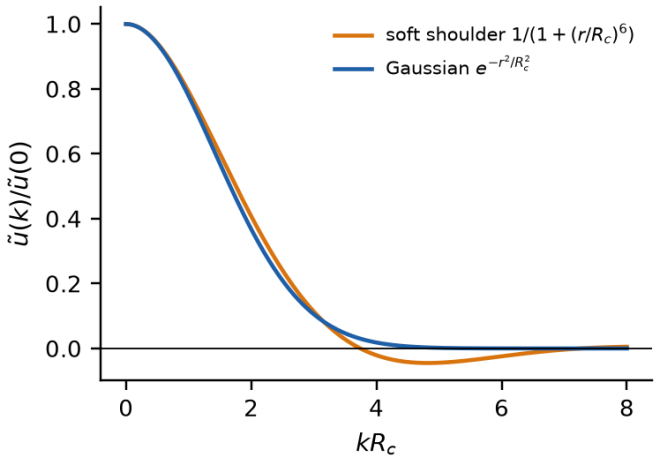


Figure 4: Normalised 2D Fourier transforms: soft shoulder $1/(1+(r/R_c)^6)$ (first negative at $kR_c = 3.75$) and Gaussian (positive). Tier X, from the same quadrature that reproduces the Gaussian closed form to 8.9×10^{-16}.

5 The rusty-SUNDIALS SCHA module

The SCHA is implemented twice. The Python class `SCHA` (`B1/code/b1_lib.py` in the results tree) was written first; the Rust module `scha.rs` (307 lines, crate `qf-yukawa-crystal`) is a second implementation that shares only the Gauss–Legendre nodes: it uses a real-space dynamical matrix and its own zone rule, and is exposed as `lattice_phonon scha a x y lam n`. It is commit `9e85e15b23be0c74a665b6a5b74ec02b0bbffc5f` on branch `physics/p2-zero-point-optimality-2026-10-25` of rusty-SUNDIALS (not pushed at the time of writing; the commit and the blob hash of `scha.rs` were read from the git object database). The module has three unit tests (harmonic reference agrees with the Poisson zone rule; small-$\hbar_{\rm eff}$ limit with $\hbar^2_{\rm eff}$ remainder; square lattice rejected and large-$\hbar_{\rm eff}$ breakdown). At paper stage we reran the crate's tests (`cargo test -offline -locked -release -p qf-yukawa-crystal`): 63 passed, 0 failed, 1 ignored (`ledger/rust_test.log`), and cross-checked Python against the Rust binary on 5 shapes: maximum $|E_{\rm py} - E_{\rm rust}| = 8.9 \times 10^{-16}$ (`data/crosscheck.json`; e.g. $E_{\rm SCHA}(a = 1, \rho, \hbar_{\rm eff} = 0.1) = 0.2267498988201642$ in both). Clippy and formatting results quoted by the producer were not rerun. This is an improvement in that the SCHA verdict no longer depends on one code, but both codes encode the same equations (Step 7), so a common conceptual error would not be caught. [X, B1-X-0012]

6 Verdict against the preregistration

item	prior	outcome
B1a harmonic $\hbar^*_{\rm eff}(a)$	0.85	Confirmed (X). Spinodal exists on $(0.37668, 1.58148)$, transition first order, winner on the harmonic stability edge.
B1b before melting	0.4	Not supported (X): $\hbar^*_{\rm eff}$ at or beyond the Lindemann band for all a; $\Lambda_1 \le \Lambda_L(\gamma_c)$ only for $\gamma_c \ge \gamma(\Lambda_1)$ and $a \approx 0.5$–1.
B1c survives beyond harmonic	0.25	Refuted within SCHA (X): none of the sampled stable lattices is below triangular up to $\hbar_{\rm eff} \approx 0.27$–$0.33$ ($\gamma_{\rm SCHA} \ge 0.60$); PIMC blocked.
B1d relevance	–	Rydberg dressing is a soft shoulder (C, quote); no cluster preemption for a pure Gaussian; quantum melting preempts.

The kill criterion is met ([X, B1-X-0013]): the transition "disappears beyond harmonic" (SCHA), and the harmonic spinodal lies at or beyond melting; the first-order point lies beyond melting except in a window of a and γ_c. The harmonic prediction is an artefact of the harmonic approximation in the sense that it does not translate into a structural transition of the quantum crystal within the theories we could run.

7 Limitations

- **SCHA is not exact** [C, B1-C-0006]. It is a Gaussian variational theory and, for each shape, a Rayleigh–Ritz upper bound on the true ground-state energy. Differences of upper bounds do not order the true energies, so "SCHA removes the transition" is a statement about the approximate landscape. Its $O(\hbar_{\text{eff}}^2)$ energy contains the quartic anharmonicity at first order but not the second-order cubic term, and the SCHA stiffening we find is itself an $O(\hbar_{\text{eff}}^2)$ effect (coefficient ≈ 14.7 at $a = 1$), exactly the order at which SCHA is incomplete. It describes neither exchange nor melting. Hence B1c holds within SCHA; the net negative verdict also rests on B1b.
- **Lindemann is a heuristic**, γ_c carries the γ_c^2 sensitivity of Step 6, the QMC anchors are for other potentials, and our reading of a_L is an assumption (Section 3.3).
- **Only Bravais lattices** with one particle per cell: non-Bravais, striped, cluster or liquid phases are not part of the energy comparison. The competitors are held in shape by an imposed constraint (they are not stress-free), and the harmonic Λ_1 winners sit on the harmonic stability edge, so harmonic theory does not identify the phase beyond Λ_1.
- **Finite sampling.** Λ_1 is a minimum over a 0.005 grid with a stability screen on 48^2 nodes plus long-wave directions; the best 12 shapes were re-verified. A coarser stability screen or a finer search could move Λ_1.
- **PIMC not run**; exchange is absent from SCHA. **No audit**: unaudited by human or AI; the same pipeline wrote the code, the numbers and this text. **Literature:** four sources re-read; the rest ([sub]) is unchecked; Likos *et al.* (cluster criterion) was not opened.
- Statements about the harmonic functional (P2) are Tier X there as well; nothing in this paper is formally verified.

8 Reproducibility and ledger

Inputs. Results copied from `results/proposal2/2026-10-26/B1/` into `data/` (`copy_data.py`, hashes in `data/MANIFEST.json`); producer code in `results/proposal2/2026-10-26/B1/code/` (commands in `TRANSITION.md`). **Paper stage:** `checks.py` (band readings, ray slopes, soft-shoulder transform, Rydberg arithmetic) → `data/extra_checks.json`; `gen_inputs.py` writes `gen_numbers.tex` and `gen/*.tex`, from which the tables and headline numbers are generated (a few values are quoted from the producer's record or derived inline, and are marked where they appear); `make_figures.py` writes `fig/`; `ledger/run_citation.py`, `ledger/scha_crosscheck.py`, `ledger/verify_refs.py`; `ledger/build_ledger.py` and `ledger/run_gate.py`; `build_pdf.py`. Python through the repository venv with `-I`. **Gate.** The Elenchus ledger gate (`tools/ledger.py`, with the evidence directory) is run on `ledger/ledger.json`; its verbatim output and exit code are in `ledger/gate_output.txt`: exit code 0; 0 blocking finding(s) and 0 flag(s) (the gate's own report; a clean run means the bookkeeping holds, not that any claim is true).

References

[1] X. Callens, *Does the triangular lattice minimise the harmonic zero-point energy? A numerical counterexample for completely monotone potentials, with kernel-checked curvature identities*, preprint v1, Zenodo, doi:10.5281/zenodo.23653833 (2026).

[2] X. Callens, *The Triangular Lattice and Universal Optimality: human proofs, Lean 4 verification, numerical validation in Python and Rust, and physics from fluids to plasmas (a textbook for students)*, Zenodo, doi:10.5281/zenodo.23295142 (2026); section on quantum melting (Lindemann conventions).

[3] G. E. Astrakharchik, J. Boronat, I. L. Kurbakov, Yu. E. Lozovik, Quantum phase transition in a two-dimensional system of dipoles, Phys. Rev. Lett. **98**, 060405 (2007); arXiv:cond-mat/0608246v2. (read, quoted)

[4] S. Prestipino, A. Sergi, E. Bruno, Freezing of soft-core bosons at zero temperature: a variational theory, Phys. Rev. B **98**, 104104 (2018); arXiv:1809.07515v1. (read, quoted)

Table 7: Ledger rows (all Tier C or X; Tier C rows depend only on Tier C rows, Tier X rows on C or X rows).

id	tier	claim (abridged)
B1-C-0001	C	model, units, E = E_st + hbar E_zp
B1-C-0002	C	isotropic curvature; spinodal hbar* = -c_st/c_zp
B1-C-0003	C	D3 cubic invariant: first order; equal and opposite ray slopes
B1-C-0004	C	Lindemann: Lambda_L ∼ gamma_c^2; ratio identity
B1-C-0005	C	SCHA closed form; small-hbar limit
B1-C-0006	C	SCHA scope: upper bound, no 2nd-order cubic, no exchange
B1-C-0007	C	Gaussian Fourier transform is positive
B1-C-0008	C	no cluster preemption for a pure Gaussian
B1-C-0009	C	Rydberg dressing: soft shoulder, not Gaussian
B1-C-0011	C	assumption: a_L = a_nn (not found in the source text)
B1-C-0101	C	quote, Astrakharchik et al. p.3: Lindemann ratio 0.230(6) at 2D dipole melting; others 0.235(15), 0.24(1), 0.279
B1-C-0102	C	quote, Prestipino et al. p.11: Gaussian (n = 2) is a marginal case, no cluster crystal in 2D or 3D
B1-C-0103	C	quote, Prestipino et al. p.15: for n = 2 crystallization is swept away at T = 0 (mean field)
B1-C-0104	C	quote, Prestipino et al. p.25: freezing completely washed out for n = 2 (mean field)
B1-C-0105	C	quote, Prestipino et al. p.25: mean field realistic for weak interparticle forces
B1-C-0106	C	quote, Henkel et al. p.2: dressed potential C6/(r^6 + Rc^6)
B1-C-0107	C	quote, Henkel et al. p.2: negative components of the momentum-space potential near 2 pi/Rc
B1-C-0108	C	quote, Henkel et al. p.4 note 26: C6 M/(hbar^2 Rc^4) < 1e-2 for their Fig. 3c
B1-C-0109	C	quote, Zeiher et al. p.2: U_ij = U0/(1 + (R/Rc)^6)
B1-C-0110	C	quote, Zeiher et al. p.3: U0/2 pi about 1.8 kHz
B1-C-0111	C	quote, Zeiher et al. p.7: Rc/a_lat = 2.02
B1-C-0112	C	quote, Zeiher et al. p.3: a_lat = 532 nm
B1-C-0113	C	quote, Zeiher et al. p.3: rubidium-87
B1-C-0114	C	quote, Zeiher et al. p.5: dressed-atom loss time 130(20) us vs 2.2 ms expected
B1-C-0115	C	quote, Zeiher et al. p.3: bare Rydberg state lifetime about 27 us
B1-X-0001	X	B1a: hbar*(a) table; negative controls a = 0.3, 2
B1-X-0002	X	first order: Lambda_1 < hbar*; stability-edge winners
B1-X-0003	X	ray slopes equal and opposite (ratio -1.0)
B1-X-0004	X	B1b: gamma at hbar* and Lambda_1 vs band
B1-X-0005	X	B1c: SCHA keeps triangular a local minimum to hbar ∼ 0.27-0.33
B1-X-0006	X	SCHA competitors incl. beyond the harmonic edge: none below
B1-X-0007	X	SCHA controls: limit, variational minimum, slope, py = rust
B1-X-0008	X	harmonic positive and negative controls
B1-X-0009	X	PIMC blocked: feasibility numbers
B1-X-0010	X	soft-shoulder Fourier transform negative; Gaussian control
B1-X-0011	X	Rydberg parameter hbar/sqrt(m U0 Rc^2) = 0.237 (arithmetic)
B1-X-0012	X	Rust SCHA module: commit, tests, cross-check
B1-X-0013	X	net verdict: kill criterion met (negative result)

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